

WARM-GLOW GIVING: EARNED MONEY AND THE OPTION TO TAKE

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Giving in dictator games has been shown to vary with the nature of the endowment (earned vs. house money) and the action space (give only vs. the option to give or take). This article is the first to test if these factors similarly affect warm-glow giving alone. There is no reason that one would expect the same outcomes given that the motivations for warm-glow giving are different from the motivations for total (warm-glow plus purely altruistic) giving. We find that warm-glow giving to charity or philanthropic institutions in a real-donation experiment increases when the endowment is earned. The option to take does reduce warm-glow giving to charity, but significant giving remains. Our results suggest that donating earned income creates greater utility than donating an equal amount from a windfall gain, and that warm glow comes not merely from the act of giving, but also from the characteristics of the recipient. (JEL C90, D64, H41)

1. INTRODUCTION

What motivates giving? A pure altruist gives solely out of concern for the well-being of recipients; giving makes the giver better off because it has made the recipient better off. Caring only that the recipient is cared for, how and by whom the care is provided is irrelevant to the pure altruist. For a pure altruist, a dollar of charitable contribution is completely crowded out by giving by any other third party. The act of giving for a pure warm-glow giver is what provides satisfaction (Andreoni 1989, 1990). A pure warm-glow giver does not care about the well-being of recipients; giving is purely egotistical.¹ Even if the recipient is not benefited, it makes the giver feel better

about herself; it increases her self-esteem, it garners praise and prestige, it satisfies a social norm.

It is the warm-glow motivation that is likely relevant for giving in large economies. Ribar and Wilhelm (2002) show that under plausible assumptions, as the size of a population increases, the marginal effect of one's contribution on the aggregate provision of a public good approaches zero. As a result, only the warm-glow motivation for giving to a public good remains. Many aspects of charitable giving are consistent with a model of warm-glow giving, including the number of individuals who give to charity, the amounts given to charity, and the non-neutrality of voluntary donations to government donations financed by taxation (Andreoni 1989, 1990, 2006; see also Cornes and Sandler 1984). While "total giving" (i.e., purely altruistic giving plus warm-glow giving) has been extensively studied, the same is not true for the warm-glow component. The potential importance of warm-glow giving for charitable giving argues for a better understanding of warm-glow behavior.

The Dictator Game has been widely used to study the extent to which individuals behave altruistically toward others.² In general, the

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1. As an illustrative example, CBS Sunday Morning reported that 67,000 teddy bears were donated to Newtown, Connecticut after a gunman killed 20 children and 6 adults at Sandy Hook Elementary School. A resident of the town stated, "I think a lot of the stuff that came into the warehouse

was more for the people that sent it, than it was for the people in Newtown" (Simon 2016).

2. The literature on dictator giving is vast. Recent contributions include Brock, Lange, and Ozbay (2013), Dreber, Fudenberg, and Randvid (2014), Dreber et al. (2013), Korenok, Millner, and Razzolini (2013), Schulz et al. (2014), Stoop (2014), and Tonin and Vlassopoulos (2014). A good, but not exhaustive, overview is provided by Vesterlund (2006).

giving measured is total giving. Two findings are of interest to this study. The first is Cherry, Frykblom, and Shogren (2002) and List's (2007) finding that dictators give less from earned endowments than from house money endowments (for other examples, see Burrows and Loomes 1994 and Hoffman and Spitzer 1985). The second is Bardsley (2008) and List's (2007) finding of fewer dictators giving and less generous giving when the action space is expanded to include the possibility of taking from the recipient.³

The contribution of this article is that it is the first to test if these two findings hold for warm-glow giving alone. There is no reason that one would expect the same outcomes given that what motivates warm-glow giving is different from what motivates purely altruistic giving (see Andreoni 1989, 1990).

Crumpler and Grossman (2008) provide a direct test of warm-glow giving in a laboratory experiment in which own-money motivated participants have no incentive to contribute money to a charity. Any contribution to charity was known to be offset by an equal reduction of the charity's endowment by the experimenter. A majority of participants gave to charity, and on average gave a significant portion of their endowment. Using the Crumpler and Grossman (2008; henceforth CG) design for testing for warm-glow giving, we examine the relative effect of earned versus house money endowments on warm-glow giving with two decision frames: one in which participants can only give to charity and a second with a symmetric decision space in which participants can either give to or take from a charity.

Our major findings are as follows. First and most novel is how earned endowments affect warm-glow giving. When the recipient is a charity, we observe larger warm-glow contributions with earned money than with house money. Indeed, significantly more subjects contribute, and mean contributions to charity are significantly higher, when the endowment is earned. This result is consistent with the interpretation provided by Cherry et al., that dictator bargaining is "more self-interested" when endowments are earned relative to house money, but this result

demonstrates that this self-interest can include the warm glow from charitable giving.

Our second major finding extends the results of Bardsley (2008) and List (2007). When the action space is expanded to permit taking, warm-glow giving is reduced, though giving does not "vanish" as in List (2007).

It should be noted that our results are consistent with the model of self-image or self-signaling developed by Benabou and Tirole (2006). The authors state the importance in moving beyond the simple dichotomy between purely altruistic and purely egoistic motivations for giving, and to distinguish, within egoistic (or warm-glow) giving, a fixed preference for giving and factors that relate behavior to self-image. Benabou and Tirole contend that these factors are context dependent. Our results identify important elements that affect how giving is perceived by the donor.

II. RELATED LITERATURE

An ongoing debate concerns the interpretation of giving in dictator experiments (Camerer 2015; Levitt and List 2007; and List 2007). Some authors have argued that the standard action space, in which the dictator can only give, engenders on the part of the dictator the expectation that a contribution should be made. This expectation is further enhanced by the fact that the experimenter, the authority figure in the rooms, has provided the dictator with the endowment ("house money") from which any contribution will be made. Finally, the expectation is even further enhanced by the experimenter asking the dictator if she would like to make a contribution to the recipient. In this context, it is questionable if any contribution can be truly interpreted as a selfless action on the part of the dictator.

Cherry, Frykblom, and Shogren (2002) and List (2007) consider the house money effect on giving. Dictators in the Cherry et al. study earned money by successfully answering quiz questions prior to playing the dictator game. These participants are paired with subjects who did not participate in the quiz activity. List has both participants earn their endowments, but the dictator earns a larger endowment than the second participant. In both studies, dictators who earn their endowments are significantly less generous and approximately three times more likely to make a \$0 offer than dictators in the respective house money treatments.

3. Eckel and Grossman (1996) report that dictators are less generous to other student subjects than to recipients (a charity) known to be more in need than the dictators. We investigate this effect in supplemental treatments reported in Appendix S2, Supporting Information (<http://ssrn.com/abstract=2791025>).

To address the action space question, Bardsley (2008) and List (2007) adapt the basic dictator experiment. They give the passive second participant his own endowment and permit the dictator to either give or take from the second participant. Both researchers find that expanding the action space significantly reduces, and can even eliminate, dictator giving.

None of these studies, however, use the real donation design where preferences for giving to a public good or charity could be expressed. Matching dictators with charities increases the parallelism between the laboratory and the field (Eckel and Grossman 1996, henceforth EG; and Eckel, Grossman, and Johnston 2005). In the context of a real-donation experiment, preferences that motivate charitable giving in practice are more likely to be active compared to allocating money to an anonymous participant.⁴

The dictator experiment has also been employed to directly test the warm-glow giving hypothesis. Such tests often use the “real donation” design of EG, in which each participant is matched with a charity of her choosing. For example, Eckel, Grossman, and Johnston (2005) do so by varying the degree of fiscal illusion in a real donation dictator experiment. The results are consistent with warm-glow giving. CG isolate the warm-glow giving in a real donation experiment by completely crowding out donations made by participants. They report that participants donated 20% of their endowments on average, and 57% of the participants made a donation. Gangadharan, Grossman, and Jones (2014) sequence the CG game with a standard real donation dictator game to categorize givers by type: nongivers, pure warm-glow givers, pure altruists, and impure altruists. They find substantial numbers in each category, as well as warm-glow giving results consistent with those reported by CG.

III. EXPERIMENTAL DESIGN AND PROCEDURE

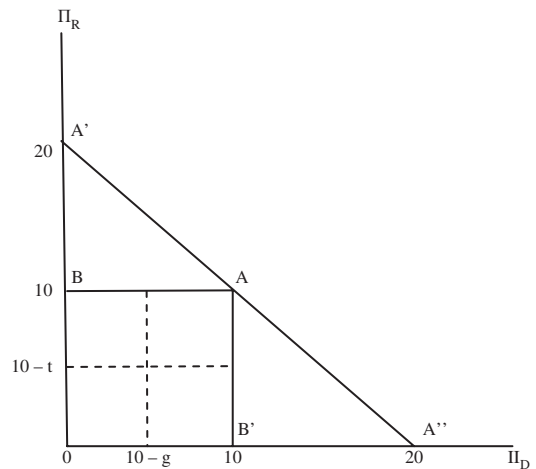
Our basic design combines elements of CG and List (2007). Each participant is endowed with \$10US, and a charity, selected by the participant, is also endowed with \$10. The participant can pass any amount from her endowment to the selected charity with the understanding that any amount passed will result in an equal reduction

TABLE 1
Treatment Characteristics

Source of endowment	House money Earned endowment	Action Space	
		Give Only	Give or Take
Treatment 1	Treatment 2	Treatment 3	Treatment 4

FIGURE 1

Payoff Possibilities for the Standard Dictator Game, and the CG Game with and without the Option to Take



in the charity’s endowment by the experimenter. The fact that the charity will receive \$10 regardless of the allocation decision by the dictator is common information.⁵ The altruistic motivation to give to charity is removed because passing money to charity does not increase the amount received by the charity; only the warm-glow motivation for giving remains.

The principal treatments involve a 2 by 2 design. We vary the source of the endowment as either house money or earned. We also vary the action set between either only giving to charity or giving to or taking from charity (see Table 1 for the design summary).

Figure 1 depicts and contrasts the action spaces for a standard dictator game and the *give*

4. EG report that giving to charity was, on average, three times the giving to the anonymous participants. We find similar results (Appendix S2, <http://ssrn.com/abstract=2791025>).

5. All participants successfully completed a questionnaire indicating that the participants understood how their allocation decision would affect both own earnings and the amount received by charity.

only and give or take treatments. The figure illustrates how the amount given affects the dictator's earnings (Π_D) and the recipient's earnings (Π_R). Let the initial endowment allocation be designated as point A. For the standard dictator game, the action space is defined by segment AA' for a *give only* treatment, and A'AA'' for a *give or take* treatment. The slope of each segment is -1 . For the *give only* warm-glow treatment, the action space is segment AB. Denote the amount the dictator passes to her selected charity as g . Passing money to the selected charity reduces the dictator's earnings by g . Although passing money to charity does not increase the charity's earnings, it does affect the perceived composition of the charity's earnings: the amount donated by the experimenter is $\$10 - g$, and the amount donated by the participant is equal to g . In the *give or take* treatment, the action space is segment AB if the dictator gives and AB' if the dictator takes. Denote the amount the dictator takes from his selected charity as t . Taking money from the selected charity reduces the charity's earnings by t . Although taking money from the charity does not increase the dictator's earnings, it does alter the perceived composition of the dictator's earnings: the amount (earned or received from the experimenter) is $\$10 - t$, and the amount received from the charity is equal to t .

We modify the CG design by adding a real-effort task before the real donation experiment. We vary whether the real-effort task earns the participants \$3, with the \$10 endowment provided by the experimenter (the *house money* treatment; 1 and 2 in Table 1), or \$13, with \$10 of this endowment used in the dictator allocation decision (the *earned endowment* treatment: 3 and 4).⁶ We also vary the feasible actions between giving only (*give only* treatment: 1 and 3) and either giving or taking (*give or take* treatment: 2 and 4). In all treatments, the charity's \$10 endowment is bestowed by the experimenter.

All sessions followed a standard three-part procedure designed to maintain comparability and to ensure participant anonymity. Consent forms were read to and signed by participants. Packets containing written instructions and slips

of paper printed with random, five-digit code numbers were distributed. An urn containing the five-digit code numbers was shown to participants, and one participant drew a number from an urn containing the five-digit numbers. The number designated the monitor who observed and assisted in conducting the experiment. The monitor received a flat fee of \$15.

Part 1 of the experiment is the real effort task. The task was designed to require all participants to exert some effort but simple enough to ensure that all participants successfully completed it.⁷ Participants were asked to count the number of zeros in a row of numbers. Successfully completing the task involved correctly counting all 15 rows. The experimenter marked any mistakes and the participant could attempt the incorrect rows again. Successful completion of the task resulted in earning \$3 for oneself (*house money* treatments: 1 and 2) or \$13 for oneself (*earned endowment* treatments: 3 and 4).

Instructions for Part 2 of the experiment were then read. The instructions informed participants in the *give only* treatment that any amount passed to the charities would reduce their own earnings and any amount passed to the charities would be offset by an equal reduction by the experimenter so that the selected charities would receive neither more nor less than \$10. In the *give or take* treatment, the instructions informed participants of the effects of passing money to the charities as in the *give only* treatment. Additionally, participants were informed that money taken from their selected charities would reduce the amount received by the charities although their own earnings would not increase.

Applying Charness and Rabin's (2002) topology of social preferences, players exhibiting narrow self-interest might give, take, or neither. They would be inclined to give if they were Andreoni egoists; they would not give if warm glow did not enter their utility functions. If such players were motivated by malice toward a charity, that malice could be acted on by selecting that charity and then taking from it.⁸ Players exhibiting competitive preferences would be disinclined to give but might take if the desire to do better than the other party was sufficiently strong and the concern for the other party was sufficiently weak. Players with difference aversion or social welfare preferences would neither give nor take.

7. Eighty-one percent of the participants successfully completed the task on the first try.

8. Charness and Rabin (2002) do not find many people to be motivated by this kind of preference.

6. In a supplemental treatment, we excluded the preallocation task in a *give-only* environment, to determine if the task itself affects behavior. This treatment also provides a replication of the CG results. The rates and amounts of giving are similar with and without the real effort task. We conclude the task does not affect behavior, and therefore our results are comparable to CG. See Appendix S2 (<http://ssrn.com/abstract=2791025>) for these results.

After reading the instructions, a questionnaire was administered to ensure participants knew how their choices would affect their own earnings and the amounts received by their selected charities.⁹ Once all questionnaires were answered correctly, the charity selection sheets were distributed (see Appendix S1, <http://ssrn.com/abstract=2791025>, for the instructions and decision sheets for the *house money/give only* and *house money/give or take* treatments). The charity selection sheet is a list of ten charities with a brief description of the services provided by the charities. After reading a summary of the instructions for Part 2, each participant selected one charity. This charity received \$10 from the experimenter. At the bottom of the sheet, a participant indicated how she wished to allocate her \$10, or if applicable, how much of the \$10 to take from charity. The instruction made clear that giving to the charity would not increase the amount the charity received, and that taking money would not increase the participant's earnings.

While earnings were tabulated, participants were given a survey to complete as Part 3 of the experiment. The survey collected demographic information, and also contained manipulation check questions as well as the open-ended questions "Please indicate what you believe the experimenters were testing for and explain what motivated you to make the decision you made" and "Please indicate what you intend to do with the money you have earned in today's experiment in general terms." A few participants (3 of 102) indicated they would give a portion of their earnings to different charities. No participants connected the effort task to the allocation decision, and no participants speculated on the design of other treatments.

All participants were invited to witness the mailing of checks to charities. Earnings envelopes with the five-digit code numbers were placed on a table near the front of the room, and participants collected their envelopes as they exited the room. The donations to the charities were totaled, checks were written and inserted in preaddressed and stamped envelopes, and the monitor and experimenter walked to the nearest mailbox and posted the envelopes. Responses to survey questions suggest that participants believed that the charities would receive

TABLE 2
Mean Response to Manipulation Check Questions

Question	Give Only Treatments	Give or Take Treatments
The procedure followed in this experiment preserved your anonymity.	4.575	4.756
The instructions for the experiment were clear and easy to follow.	4.475	4.105
The charities selected will receive a donation from the experimenter.	4.733	4.733
The charity you selected efficiently uses donations for their stated cause.	4.542	4.465

the donations, that their own decisions were anonymous, and that the instructions were clear. Using a Likert scale from 1 (*strongly disagree*) to 5 (*strongly agree*), the mean response to each of these questions is more than 4 (see Table 2).

IV. RESULTS

Participants were recruited from undergraduate classes at Mississippi State University. One hundred and two participants took part in the experiment, excluding those selected as monitors.¹⁰ Five sessions were conducted. Treatment 1 consists of two sessions of 17 and 10 participants each; treatments 2, 3, and 4 were conducted in single sessions of 25, 27, and 23, respectively. Each session lasted between 45 minutes and 1 hour. Table 3 provides summary statistics by treatment.

We begin by testing the null hypothesis that, for all treatments, the distributions of choices are drawn from parent populations with the same arithmetic mean. A Kruskal-Wallis test rejects this null hypothesis at the 10% level when correcting for ties ($X^2 = 7.88$, $p = 0.09$), suggesting that at least one sample distribution is significantly different.

To determine which treatments differ, we conduct pairwise comparisons of the distribution of contributions using the Wilcoxon rank-sum test. The null hypothesis is that each pair of samples is drawn from populations with the same distribution. Table 4 displays the Wilcoxon test

9. Subjects who responded incorrectly to the questionnaire were counseled how to correctly determine own and charity's earnings, and were given additional opportunities to answer correctly.

10. An additional 91 participants took part in supplemental treatments referenced in Appendix S2 (<http://ssrn.com/abstract=2791025>).

TABLE 3
Summary Statistics by Treatment

Treatment	Primary Treatments			
	1	2	3	4
Feasible actions	Give only	Give or take	Give only	Give or take
Source of endowment	House money	House money	Earned endowment	Earned endowment
% Donating	40.7%	20.0%	59.3%	43.5%
% Taking	n/a	0	n/a	0
Mean donation (unconditional)	\$1.52	\$0.60	\$2.41	\$2.09
Standard deviation	2.55	1.44	3.18	3.15
Mean donation (conditional)	\$3.73	\$3.00	\$4.06	\$4.80
<i>n</i>	27	25	27	23

Note: n/a, not applicable.

TABLE 4
Comparisons of Giving: Two-Sample Wilcoxon Rank-Sum Tests

	House Money/ Give Only	Earned Endowment/ Give Only	House Money/ Give or Take
Earned endowment/give only	1.23 (0.22)	—	—
House money/give or take	−1.62 (0.11)	−2.81*** (0.01)	—
Earned endowment/give or take	0.49 (0.62)	−0.68 (0.50)	1.91* (0.06)

Note: Test statistic z , with $\text{pr} > |z|$ in parentheses.

* $\text{pr} |z| < 10\%$; *** $\text{pr} |z| < 1\%$.

results. For each decision frame, we do not observe lower, but on the contrary, higher contributions when endowments are earned. In the *give only* treatments, the mean donation is \$1.52 in the *house money* treatment and \$2.41 in the *earned endowment* treatment. In the *give or take* treatments, the mean donation is \$0.60 in the *house money* treatment and \$2.09 in the *earned endowment* treatment. We now formally test our null hypotheses.

A. The Effect of Earned and House Money Endowments on the Number of Warm-Glow Givers and the Amount of Warm-Glow Giving (Treatments 1 vs. 3; 2 vs. 4)

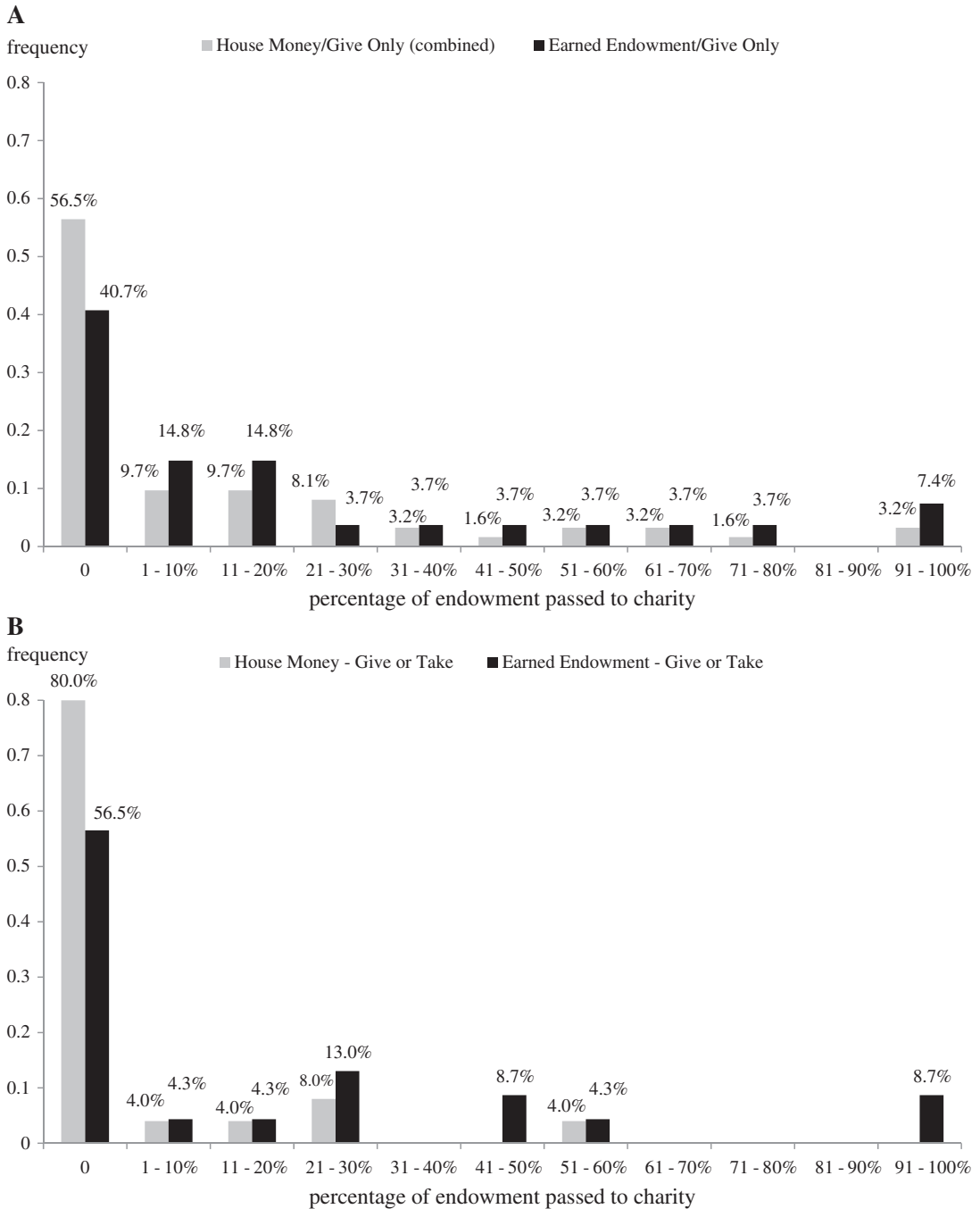
HYPOTHESIS 1: *The proportion of givers and the average amount given will decline when endowments are earned rather than house money.*

RESULT 1. *In both the give only and the give or take treatments, we find the proportion of dictators does not decrease and may even increase. Earned endowments are associated with equal or increased giving in both decision frames.*

Support. In stark contrast with Cherry, Frykblom, and Shogren (2002) and List (2007),

we do not find a decrease in the proportion of dictators giving in the *earned endowment* treatment relative to the *house money* treatment. Figure 2A displays the histogram of choices in the *give only* treatments; Figure 2B displays this information for the *give or take* treatments. Visual inspection of these figures suggests that, for a given decision space, giving not only persists, but actually increases, when participants earn the endowment. In our *give only/house money* treatment, 40.7% (11 of 27) pass money; in the *give only/earned endowment* treatment, 59.3% (16 of 27) pass money. The difference is not statistically significant ($\chi^2(1) = 1.85$, $p = .17$). Nor are the distributions significantly different (Kolmogorov-Smirnov test statistic $D = 0.19$, $p = .65$). Although the nonparametric Kolmogorov-Smirnov test cannot distinguish between the distribution of giving in the *give or take* treatments ($D = 0.23$, $p = .42$), in the *give or take/house money* treatment, 20% (5 of 25) pass money; in the *give or take/earned endowment* treatment 44% (10 of 23) pass money. This difference is significant ($\chi^2(1) = 2.08$, $p = .08$). Furthermore, none of the participants took from the charity when they had the option to do so.

FIGURE 2
Distribution of Donations to Charity as a Percentage of the Endowment



Note: (A) Give only, house money and earned endowment treatments. (B) Give or take treatments, house money and earned endowment treatments.

TABLE 5
Giving to Charity: Probit Model Results

Demographic Characteristics	1 No	2 Yes
Earned endowment	0.560*** (0.060)	0.578*** (0.032)
Take	-0.498*** (0.060)	-0.457*** (0.016)
Female		-0.377 (0.295)
Non-White		1.025*** (0.138)
Constant	-0.280*** (0.048)	-0.856*** (0.137)
Log-likelihood	-64.836	-58.968
Observations	102	102

Notes: The dependent variable is equal to 1 if the participant passed a positive amount to charity. Robust standard errors are in parentheses. Models 1 and 2 cluster standard errors on the treatment level.

***Significant at 1% level.

While the differences are not significant or only weakly significant, the direction of the change is the opposite of the significant negative change reported by Cherry et al. and List for total (warm-glow plus altruistic) giving. It must be remembered that our comparison is not really to a null hypothesis of no effect, but rather to the findings of Cherry et al. and List that earned endowments significantly decrease the number of givers and the amount of giving. Our finding that earned endowments leave unchanged or increase both the number of warm-glow givers and the amount of warm-glow giving, still raises important issues with the Cherry et al. and List findings.

Table 5 presents results from probit models. Models 1 and 2 decompose the treatment effects by the source of the endowment (*Earned Endowment* equals 1 if the endowment is earned) and the option to take (*Take* equals 1 if participants have the option to take). Model 1 controls only for the source of the endowment and the option to take; Model 2 additionally controls for gender and race. The dependent variable equals 1 if the participant passed a positive amount to a charity and zero otherwise. *Earned Endowment* equals 1 if the endowment is earned, and *Take* equals 1 if participants have the option to take. Results from Models 1 and 2 indicate that participants are significantly more likely to give when the endowment is earned.¹¹ Expanding the decision set to include taking does reduce giving, yet no participant took money from a charity. We discuss the effect of the decision space in more detail in subsection B below.

11. Logit models yield similar results.

B. The Effect of the Action Space on the Number of Warm-Glow Givers and the Amount of Warm-Glow Giving

HYPOTHESIS 2: *The proportion of giving dictators and the average amount given will decline when the action space is expanded to permit taking.*

RESULT 2. *An expanded action space is associated with a reduction in the proportion of giving dictators, and a decrease in the average amount given.*

Support. In the *house money* treatments, dictators passed on average \$1.52 to charity in the *give only* treatment and \$0.60 in the *give or take* treatment. This difference approaches statistical significance, but is not significant at conventional levels with the Wilcoxon rank-sum test (see Table 4). The effect of the action space is more muted with earned endowments. With earned endowments, dictators passed on average \$2.41 in the *give only* treatment and \$2.09 in the *give or take* treatment, an insignificant difference (see again Table 4). Fifty percent (27 of 54) of the dictators in *give only* treatment passed a positive amount; 31% (15 of 48) of the dictators in the *give or take* treatment passed a positive amount. This difference is significant ($\chi^2(1) = 3.69$, $p = .06$). This result is supported by probit results reported in Table 5. In both models, we find that the expanded action space significantly reduces the probability that a dictator will pass a positive amount to charity. This finding is consistent with the Bardsley (2008) and List (2007) finding for total giving.

Additional insight for both endowment and action space effects is provided by examining the determinants and likelihood of giving. Because data are censored at the upper and lower limits, we run tobit models with the amount passed to charity as the dependent variable. Results are shown in Table 6.

Models 1 and 2 decompose the treatment effects by the source of the endowment (*Earned Endowment* equals 1 if the endowment is earned) and the option to take (*Take* equals 1 if participants have the option to take). Model 1 controls only for the source of the endowment and the option to take; Model 2 additionally controls for gender and race. Both models indicate that donations are significantly higher with earned own endowments relative to the house money sessions, and the option to take significantly reduces donations.

TABLE 6
Amount Given: Tobit Model Results

Demographic Variables	(1) No	(2) Yes
Earned Endowment	3.262** (0.752)	3.084*** (0.420)
Take	-2.249*** (0.586)	-2.161*** (0.357)
Female		-1.834 (1.818)
Non-White		2.823*** (1.045)
Constant	-2.014*** (0.676)	-3.930* (1.211)
Log-likelihood	-158.202	-156.252
Observations	102	102

Notes: Dependent variable is amount passed to charity. Robust standard errors are in parentheses. Standard errors are clustered on the treatment level.

*Significant at 10% level; **significant at 5% level; ***significant at 1% level.

C. Supplemental Treatments

Two additional supplemental treatments were conducted to ensure internal and external validity. Participants were paired with other participants instead of a charity (one *house money*, one *earned endowment*). If observed giving is due to confusion or a demand effect in the way the decision sheet is presented, we should observe similar rates of giving, which we did not. As in EG, we find that context matters: giving is significantly reduced when participants are paired with other participants (see Appendix S2, <http://ssrn.com/abstract=2791025>).

V. SUMMARY AND CONCLUDING REMARKS

Previous work employing the Dictator Game has shown that total (warm-glow plus purely altruistic) giving is drastically reduced when participants earn their endowments in a pregame task (Cherry, Frykblom, and Shogren 2002 and List 2007), and that total giving is reduced, and considerable taking occurs, when the action space is expanded to include taking as well as giving (Bardsley 2008 and List 2007). We test if these results are robust for warm-glow giving alone. The fact that the motivations for warm-glow giving are different from the motivations for purely altruistic giving suggests that one should not necessarily expect the same outcomes for warm-glow giving as for total giving.

To test our hypotheses, we adopt the CG design. The treatments provide context by pairing dictators with charities of their own choice. A donation to charity does not increase the amount the charity receives. Instead, it alters the perceived composition of the donation to charity

between the experimenter and the participant. We employ a real effort task that either earns \$3 for the participant, separate from the \$10 used in the real donation experiment (*house money* treatment), or earns \$13 for the participant, of which \$10 is used in the real donation experiment (*earned endowment* treatment). We examine the effect of earned endowments with two decision frames. Participants can only pass money to a charity in the *give only* treatment; participants can either pass money to charity or take money from charity in the *give or take* treatments.

While Cherry, Frykblom, and Shogren (2002) and List (2007) find that earning the endowment significantly reduces dictator (total) giving, we find that earning the endowment increases warm-glow giving. We observe statistically insignificant increased warm-glow giving in our *give-only* treatments and statistically significant increased warm-glow giving in our *give or take* treatments. In our *give only* treatments, approximately 50% more participants give and they give approximately 50% more when endowments are earned rather than house money. The difference in our *give or take* treatments is even more pronounced; 100% more participants give and average giving is about 300% higher. The qualitative dependent variable and limited-dependent variable models indicate that the proportion of subjects who donate to charity and the amount given to charity are higher in treatments in which the endowment is earned. When we expand the action space, we find results consistent with Bardsley (2008) and List (2007). Allowing subjects to take as well as give reduces, but does not eliminate warm-glow giving, and we observe no taking.¹²

Our results are consistent with the hypothesis that participants receive a warm glow from giving. Although a selected charity receives \$10 regardless of the amount passed to the charity, a participant who passes \$2 to charity may feel as if her contribution accounts for \$2 of the \$10 total. It must be recognized, as the Lei, Noussair, and Plott (2001) Active Participation Hypothesis would suggest, that at least some of the giving we report is due to a desire on the part of experiment participants to do something. While our subjects in the *give or take* treatments have two action options, the taking action may be

12. EG find that context matters for a dictator game; total giving was greater when dictators were paired with a charity rather than another anonymous participant. We find that this result holds for warm-glow giving (see Appendix S2, <http://ssrn.com/abstract=2791025>).

sufficiently distasteful as to not be considered an option. However, while the Active Participation Hypothesis may explain some of the giving behavior observed, it does not explain the increased number of givers when the option to take is available.¹³

The results in this article allow us to refine the interpretation of earned endowments on dictator giving. To declare that earned endowments decrease giving is an overstatement. We do observe decreased giving in response to earned endowments in treatments when other anonymous participants are the recipients of the dictators' allocations. The results from the real donation experiments are consistent with the hypothesis that dictators make more self-interested decisions when endowments are earned. Self-interest, for a subset of dictators, includes the warm glow of donating to a charitable organization. Therefore, more self-interested behavior is not always equivalent to less giving.

Our results suggest that donating one's earned income creates greater utility than donating an equal amount of a windfall gain.¹⁴ These results are consistent with the Benabou and Tirole (2006) model of self-image. Actions allow individuals to define their self-image. However, Benabou and Tirole state the importance of delineating between a preference for warm glow and "... motives that relate to what a person's behavior says about him or her, which will depend on the informational and economic context..." The results in this article indicate that donating earned money is seen as more of a sacrifice than donating house money, and this is compounded by the social norm regarding the recipient of donated money.

13. Our findings may also understate the level of giving. Players could make donations from their earnings after the experiment and our design only targets "internal" sources of warm glow, that is, feeling good about one's self and satisfying a social norm. It does not target "external" sources such as prestige and the acclaim of others.

14. Also, warm glow comes not merely from just the act of giving, but also from the characteristics of the recipient (see Appendix S2, <http://ssrn.com/abstract=2791025>). Gangadharan, Grossman, and Jones (2014) report results consistent with ours. They find that warm-glow givers are more generous to needy Indigenous Australians than to needy non-Indigenous Australians. They argue that "... Indigenous Australians are likely to be viewed as more in need than Australians in general ... the social and economic outcomes of Indigenous Australians are significantly worse than non-Indigenous Australians."

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SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Instructions for *house money/give and house money/give or take* treatments

Appendix S2. Analysis of supplemental treatments

Table S1. Summary statistics by treatment

Figure S1. Distribution of amount passed to charity as a percentage of the endowment: *House money/Give only* treatments, without and with the real-effort task